1.3.3 ▶ ▶ ▶ ▶ Conservation of Momentum

Momentum for a system of particles is conserved when there are no external net forces acting on the system. To derive this fact, we start by considering a system of two particles that interact via some interaction force for a time Δt.

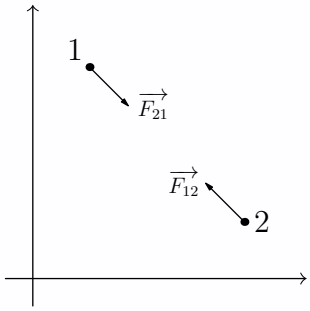

Fig. 1.5

In this diagram, $\mathbf{F_{12}}$ is the interaction force that 1 exerts on 2 while $\mathbf{F_{21}}$ is the force that 2 exerts on 1. Thus, we can calculate the impulse experienced by each particle.

$$\Delta \mathbf{p}_1 = \mathbf{F}_{21}\Delta t$$

$$\Delta \mathbf{p}_2 = \mathbf{F}_{12}\Delta t$$

Since $\mathbf{F}_{12}$ and $\mathbf{F}_{21}$ are a Newton's 3rd law pair, we have

$$\mathbf{F}_{21} = -\mathbf{F}_{12} \tag{1.45}$$

OR

$$\Delta \mathbf{p}_1 = -\Delta \mathbf{p}_2. \tag{1.46}$$

Thus, the total change in momentum of the system is

$$\sum \Delta \mathbf{p} = \Delta \mathbf{p}_1 + \Delta \mathbf{p}_2 = 0. \tag{1.47}$$

Note that we assumed that there were no external net forces to the system. If there were, the total change in momentum of each particle would not be equal in magnitude and thus momentum would not be conserved ★.